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Audience: SG1, LEWG, WG21

Merge most of Networking TS into C++ Working Draft

Introduction

The Networking TS grew out of Boost::ASIO and has got a lot of usage experience as part of Boost and in standalone implementations of the TS. Most of the Networking TS is uncontroversial.

One specific part of the Networking TS however clashes with the executor work in SG1. The executor work from SG1 is not yet ready to be merged into the C++ working paper. So this part of the Networking TS should not be merged. But most of the Networking TS can work without an explicitly specified executor model. This paper proposes to merge the executor independent parts of the Networking TS into the C++ working paper and solve any open issues on the Networking TS inside the C++ working paper.

Wording Overview

The proposal is to merge the Networking TS into the C++ working paper with the following changes:

- Move the contents of the namespace `std::experimental` to namespace `std`.
- Rename the headers `<experimental/*>` to `<*>`.
- From table 3 remove `ExecutionContext` and `Executor`.
- From header `<netfwd>` ([fwd.decl.synop]) remove:
 - `executor_binder`
 - `executor_work_guard`
 - `system_executor`
 - `executor`
 - `strand`.
- From header `<executor>` ([async]) remove all but:
 - `async_result`, but remove specializations with `use_future_t` and `packaged_task`
 - `async_completion`
 - `associated_allocator`
 - `associated_allocator_t`
 - `get_associated_allocator`
 - `fork_event`
 - `uses_allocator`.
 - Also remove the respective specifications.
- From [async.reqmts] remove:
 - [async.reqmts.executor]
 - [async.reqmts.executioncontext]
 - [async.reqmts.service]

- [async.reqmts.async.assoc.exec]
 - [async.reqmts.async.io.exec]
 - [async.reqmts.async.handler.exec]
 - [async.reqmts.async.work]
 - [async.reqmts.async.completion]
 - [async.reqmts.async.exceptions]
 - [async.reqmts.async.composed].
- From [async.reqmts.async]p1 remove Executor1, Executor2, ex1, ex2, work1 and work2.
- From class io_context ([io__context]) remove:
 - the public base class execution_context
 - executor_type
 - get_executor
 - paragraphs 1, 2, 4.1 and 4.2.
- In class io_context ([io__context]) change paragraph 6 remove get_executor, and ", and the io_context::executor_type copy constructors, member functions and comparison operators,".
- Remove [io__context.exec].
- From basic_waitable_timer ([timer.waitable]), basic_socket ([socket.basic]), basic_socket_acceptor ([socket.acceptor]), and ip::basic_resolver ([internet.resolver]) remove:
 - executor_type
 - get_executor
 - from the constructors and assignment the postcondition about get_executor
- From table 17 ([buffer.stream.reqmts.asyncreadstream]) and table 19 ([buffer.stream.reqmts.asyncwritestream]) remove the get_executor requirement.
- In basic_socket_acceptor ([socket.acceptor]) in the accept functions that don't take an io_context& as argument replace in the Returns clause get_executor().context() with io_context().